AutoRegAI: LLM-Powered Compliance Intelligence for ADAS Systems

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1、Executive Summary

1.1 Project Overview

Given that the automotive and autonomous automotive sectors are highly regulated, a vehicle must always comply with all current regulations or risk non-compliance. With laws and regulations continually evolving at both the state and national levels, ensuring that a vehicle complies with all applicable requirements at all times is challenging. However, this difficult task can be significantly easier with an LLM tailored to interpret all regulations and generate developer-understandable recommendations, which can then be implemented in the ADAS system. Overall, this LLM is expected to significantly shorten the developer timeline and reduce the risk of non-compliance, with the only major risk being that the LLM user is responsible for all implemented outputs.

Our team has created a proof-of-concept LLM that shows the type of outputs that can be achieved via a legal assistance LLM. Specifically, this LLM is trained on Pennsylvania laws and regulations surrounding L2 autonomous vehicles and is trained to answer questions like "Who is responsible if a Level 2 ADAS vehicle crashes in Pennsylvania?" and "What are the limitations of ADAS?" The final LLM's goals are to help bridge the legal knowledge gap between legal personnel and software developers, mitigate non-compliance risks, decrease the timeline from when laws are updated/published to implementation into ADAS, and more.

1.2 Achievements to Date

The project has already progressed from an initial idea to a working prototype. A complete Retrieval Augmented Generation pipeline has been built using LangChain, Chroma, and embeddings, allowing the system to search the document database and generate answers based on relevant source material. Real legal and regulatory documents have also been collected and added to the system, including Pennsylvania Title 75 and the NHTSA Standing General Order, so the model is grounded in actual compliance-related materials rather than sample data.

A working demo has also been implemented to support regulatory question answering and scenario testing. In addition, an initial set of evaluation questions has been designed to test both compliance-related questions and scenario-based reasoning. These achievements show that the project has already moved beyond the planning stage and now includes a functioning technical pipeline, a real document base, a user-facing demo, and an early evaluation framework for future testing and improvement.

1.3 Problem Context and Industry Challenges

The autonomous driving industry is currently facing:

• Complex regulations (state, federal, and evolving):

Since the autonomous driving space is ever evolving and has recently had a significant boom in popularity, regulatory bodies are constantly deploying new laws and regulations affecting autonomous vehicles (AVs) at all levels. With each state developing differing regulations for autonomous vehicles, and no federal laws, this poses a critical issue for ADAS developers on how to keep up with the every changing legal space.

Currently, the National Highway Traffic Safety Administration (NHTSA) has provided some guidelines for autonomous vehicle safety standards, but no solid laws or regulations. However, 35 states have passed laws about AVs, each with its own set of regulations (UrbanSDK). This poses a huge risk of non-compliance for AV testing, as oftentimes, tests must run across state lines, and what is compliant in one state may not be the same for its neighboring states.

• Company Liability and Extremely high compliance costs (compliance failure → recall/liability)

This non-compliance risk can result in million dollar loses for an AV company. In fact, California has already begun to implement legislation that allows law enforcement to issue "notices of noncompliance" to autonomous car companies for traffic violations (2025 Maripuu). Fines for millions of cars can add up, thus making noncompliance a critical threat.

There are nationwide increasing regulations made concerning autonomous vehicle companies' liability when something goes wrong. In instances where an autonomous vehicle is involved in a crash, liability may fall on the company or software developer when the vehicle's design, hardware, or safety procedures fail to detect another car or pedestrian, or if any part of the autonomous vehicle malfunctions or miscalculates driving conditions (2026 Marco and Nesselbush). Even though L2 autonomous vehicles are defined as requiring the human to oversee and immediately intervene with the ADAS, companies can still face lawsuits based on liability, which makes compliance of the utmost importance.

Additionally, many autonomous car companies are being held partially liable for crashes and misuse of their systems. In 2025, a Jury in Miami, Florida, determined that Tesla is held 33% liable for a fatal crash in 2019, and thus must pay $42.5 Million in compensatory damages (2025 Kolodny). Liability cases like Tesla's are not uncommon and thus pose a huge risk for AV companies.

1.4 LLM Cost

• API Integration (Integration into enterprise development workflows)

The primary cost component of deploying an LLM-based system comes from API integration. In our demo model, we use a free API provided by ChatGPT4.0 for model experimentation. While the model is cost-efficient for prototyping, it introduces limitations in terms of integration speed, scalability, and system control. In a production model, we suggest that Latitude AI adopt a paid API solution, which enables more reliable performance and seamless integration into enterprise development workflows.

We compared multiple LLM API solutions based on cost, performance, latency, and scalability to evaluate the most suitable option for enterprise deployment. Our current prototype uses the free version of GPT-4o, which is cost-efficient but limited in terms of scalability and integration.

Among the options, Gemini Flash and AWS Titan offer the lowest cost, making them attractive for large-scale deployment. OpenAI GPT-4o provides a strong balance between cost and reasoning performance, while Anthropic Claude Sonnet delivers the highest reasoning quality at a higher cost.

Overall, the trade-off between cost and performance should be aligned with the system's requirements, particularly in safety-critical applications like ADAS, where accuracy and reliability are essential.

	Option A: Current prototype	**Option B: OpenAI**	**Option C: Gemini**	**Option D: Anthropic**	**Option E: AWS**
Core Model	GPT-4o (Free)	GPT-4o (API)	Gemini Flash	Claude Sonnet	AWS Titan
Input Cost (per 1M token)	$0	$2.50	$0.1	$3	$0.3
Output Cost (per 1M token)	$0	$10.00	$0.6	$15	$0.9
Est. Monthly Cost	~ $0.00 USD	~ $15.00 USD	~ $1.00 USD	~ $18 USD	~ $1.2 USD
Performance	Low	High	Medium	Very High	Medium
Latency	Slow	Fast	Medium	Very Fast	Fast

• Enterprise Subscription (Regulatory updates + Custom models)

In addition to API usage costs, enterprises may incur ongoing subscription and maintenance costs when deploying an LLM system in a real-world setting. Key components of the costs include:

- **Regulatory Data Updates and Maintenance**: Companies need to maintain up-to-date regulatory data. Since state laws and ADAS-related regulations continuously evolve, the system must regularly update and internal legal documents in the RAG pipeline, while also maintaining version control for traceability and compliance. Without this maintenance, the model may generate outdated or incorrect requirements, leading to potential legal risks. For example, if a state updates its driver monitoring requirements but the system is not updated accordingly, the model could produce incorrect ADAS behavior recommendations.

- **Customize Model:** In addition to maintaining regulatory data, companies must continuously adapt and customize the model to ensure it performs effectively within the ADAS. This includes refining system prompts, improving output structure, and incorporating domain-specific knowledge such as ADAS behavior definitions and safety requirements. Over time, further enhancements—such as fine-tuning with

labeled examples or optimizing retrieval strategies—may be required to improve accuracy and consistency.

Overall, these enterprise-level costs are essential to ensure the system remains accurate, reliable, and compliant over time, as real-world LLM deployment requires continuous updates and refinement.

2、Product & Technology

2.1 Product Overview

2.1.1 System Features

The system is built around four main features that support legal research and compliance checking for ADAS-related questions. These features are designed to make legal materials easier to search, interpret, and apply in practical driving scenarios. The main functions of the system include Legal Q&A, Scenario Compliance, Behavioral Recommendations, and Liability Risk Flagging.

1. **Legal Q&A** allows users to ask legal or regulatory questions in natural language and receive answers based on the documents stored in the database. Instead of requiring users to read through long statutes or reports manually, the system retrieves relevant passages and uses them to generate a grounded response. This makes it easier to answer questions about Pennsylvania vehicle rules, highly automated vehicle requirements, and federal crash reporting obligations.
2. **Scenario Compliance** helps users test whether a specific driving situation appears compliant with the rules in the database. This is useful because ADAS compliance is often not just about definitions, but about how the system behaves in a real situation. For example, the system can be used to check questions involving lane use, following distance, right of way, distracted driving, or whether a crash involving Level 2 assistance may trigger reporting requirements.
3. **Behavioral Recommendations** translates legal requirements into more practical guidance for system behavior. Rather than only stating what the law says, the system can point to what an ADAS-related design should pay attention to, such as keeping the driver engaged in Level 2 settings, following traffic rules, or recognizing when a vehicle should move into a minimal risk condition in more advanced automation contexts. This feature is especially useful for connecting legal language to engineering or design decisions.
4. **Liability Risk Flagging** helps identify situations that may create higher legal risk. This does not mean the system makes a final legal judgment, but it can highlight situations where responsibility and compliance may become more important, such as crashes involving driver inattention, improper system use, reporting failures, or unclear boundaries between driver responsibility and automated system behavior. This feature is important because Level 2 systems still require the human driver to remain engaged, while other automated systems may involve different legal duties and reporting rules.

Together, these four features show that the system is not only a document search tool, but also

a practical compliance assistant. It is designed to help users find legal information, apply it to real situations, and better understand where compliance and liability risks may arise.

2.1.2 Data Collection

1. Data Sources:

<u>Academic Papers:</u>

1. *Introducing Advanced Driver Assistance Systems: Some Legal Issues*

   This paper examines the legal challenges posed by Advanced Driver Assistance Systems (ADAS), focusing on the adequacy of existing traffic safety regulations and liability frameworks. It concludes that while current laws offer some flexibility for technical innovation, there is a significant need for international safety standards to address potential system failures and the shifting responsibilities between drivers and manufacturers.

   This information is critical for our database as it provides a foundational legal framework for how ADAS technologies must interact with existing traffic safety regulations and international design standards. It highlights the critical shift in liability from the driver to the manufacturer as systems become more advanced, which is a primary legal concern for L2 developers. Furthermore, it identifies the necessity for global safety requirements, helping developers understand the regulatory landscape beyond local statutes.

2. *Product Liability for ADAS: Legal and Human Factors Perspectives*

   Wees and Brookhuis examine the European Product Liability Directive's concept of a "defective product", specifically as it applies to the development and implementation of ADAS. It explores the "consumer expectations" and "risk-benefit" tests used to determine liability, arguing that both legal approaches require integration with human factors expertise to be properly applied. Ultimately, they address how potential liability for system developers and manufacturers acts as a significant non-technical barrier to the rapid deployment of these safety technologies.

   This information is important for our database as it provides a legal framework for defining "defective" ADAS products using consumer expectations and risk-benefit analyses, which are critical for determining liability in L2 system failures. It also integrates human factors expertise, helping developers understand how the interaction between the driver and the technology influences legal responsibility and the safety-benefit balance of their systems.

3. *Car Minus Driver: Autonomous Vehicles Driving Regulation, Liability, and Policy*

   Damien A. Riehl explores the legal and regulatory shift from human-driven to autonomous vehicles, specifically addressing how existing common law, statutes, and insurance models must adapt to a world without human drivers. He details the transition from driver-assist technologies to full autonomy, analyzing the resulting impact on product liability, privacy, and the potential for a proactive legislative approach to shape the future of transportation.

This paper is essential for the database because it provides a comprehensive overview of the transition from driver-assist technologies to full autonomy, highlighting the evolving legal standards for regulation, liability, and policy. It also analyzes how existing common law and statutes must adapt to these technological shifts, which is critical for helping L2 developers navigate the changing landscape of legal responsibility and insurance models.

4. *Liability in Autonomous Vehicle Accidents*

   Pinter, Szalay, and Vida examine how the shift from human-led to autonomous driving requires new methods for accident reconstruction and the distribution of legal liability. They emphasize the critical role of Event Data Recorders (EDRs) in providing objective evidence to determine whether a crash was caused by human error, environmental factors, or technical system failure. Ultimately, the authors argue for a modernized legal framework that can distinguish between driver negligence and manufacturer product liability as vehicles take over more control.

   This document is crucial for the database as it introduces the technical necessity of Event Data Recorders (EDRs) to provide the objective data required to distinguish between driver error and system failure in L2 accidents. It provides the "forensic" perspective of liability, helping developers understand how their system's data logs will be used as primary evidence in legal reconstructions and the distribution of blame.

<u>Pennsylvania Legal Database:</u>

1. Autonomous Vehicle Laws in Pennsylvania

   As of July 2023, this law declares autonomous vehicles (AVs) such as passenger cars, commercial trucks, and delivery vehicles to be legal in Pennsylvania, but they must obtain PennDOT permission and follow strict safety rules. The law distinguishes between three levels of technology: automated vehicles (Levels 0-2), which require a human driver, highly automated vehicles (Level 3) that may require intervention, and autonomous vehicles (Levels 4-5) that operate completely independently. To operate, companies must provide detailed safety plans for testing or secure a certificate of compliance for commercial use, and all such vehicles must carry at least $1 million in liability insurance.

   This legislation is important to our database as it explicitly distinguishes between "Automated" (Levels 0-2) and "Autonomous" (Levels 4-5) vehicles, ensuring developers correctly identify that their systems require the human driver to remain legally responsible and engaged at all times. It provides specific compliance benchmarks, such as the requirement for a $1 million liability insurance policy and the mandate that human drivers must be able to take control immediately. Furthermore, it highlights the complex liability landscape in Pennsylvania, noting that developers could be held responsible for "software defects" or "flawed decision-making algorithms" under product liability law.

2. *PA Act 130 of 2022 (Formerly House Bill 2398)*

   Legislation that modernized the state's Vehicle Code to officially permit and regulate

the operation of highly automated vehicles (HAVs). The law allows for "driverless" operations on public roads, provided the vehicle is operated by a certified entity and can achieve a "minimal risk condition" (safely pulling over) in the event of a system failure. In addition to autonomous vehicle rules, the act includes "omnibus" amendments that established new criminal penalties for the theft of catalytic converters and updated requirements for reporting accidents involving automated systems. This allowed Pennsylvania to transition from a testing-only to a full deployment framework, aiming to cement the state as a national hub for self-driving technology and innovation.

This piece of legislation is important to our database as it officially codified the framework for "Highly Automated Vehicles" in Pennsylvania, which creates a legal distinction between autonomous systems and the L2 ADAS. It establishes the critical requirement that even in advanced systems, a "human driver" must be able to take immediate control, which directly informs our LLM's compliance with highway "hand-off" protocols. By incorporating this Act, the database gains the specific statutory language used to define the "Dynamic Driving Task," ensuring the LLM understands the exact legal boundary where the company's responsibility ends, and the human driver's begins under PA law.

3. *PA Title 75 (The Vehicle Code)-Chapter 33 Rules of the Road*
4. *PA Title 75 (The Vehicle Code)-Chapter 85 HAV Law*
5. *PA Title 75 (The Vehicle Code)- Chapter 35 (Special Vehicles and Pedestrians)*

Pennsylvania Title 75 (The Vehicle Code)- Chapter 33 (Rules of the Road in General), 35 (Special Vehicles and Pedestrians), and 85 (Highly Autonomous Vehicles): Chapter 33 (Rules of the Road in General) establishes the fundamental "handbook" for driving in Pennsylvania, detailing essential traffic laws such as right-of-way, speed limits, and the requirement to drive on the right side of the road. Chapter 35 (Special Vehicles and Pedestrians) outlines the specific rights and responsibilities of non-standard road users, including pedestrians, bicyclists, and motorcyclists, ensuring they can safely coexist with motorized traffic. Chapter 85 (Highly Automated Vehicles) provides the modern legal framework for self-driving technology, permitting the operation of autonomous vehicles with or without a human driver, provided they meet strict safety certification and insurance requirements.

Highlighting these specific chapters of PA Title 75 is critical to the database as they define exactly how an L2 system must behave relative to other road users on Pennsylvania highways. Chapter 33 provides the strict operational rules for lane-keeping and following distances that L2 steering and acceleration systems must replicate, while Chapter 35 mandates how these systems must legally detect and react to vulnerable road users like cyclists. Crucially, Chapter 85 establishes the legal boundary between "Highly Automated Vehicles" and Level 2 systems, ensuring the LLM correctly identifies that L2 systems remain under "human driver" liability rather than the stricter autonomous certification. Together, they provide the complete statutory framework for any company to prove its systems are "careful and prudent" under Pennsylvania's unique 2026 highway safety standards.

6. *The National Highway Traffic Safety Administration's Standing General Order on Crash Reporting*

While not a formal law, this compulsory administrative order mandates that manufacturers and operators report specific crashes involving L2-L5 vehicles. For L3 to L5 vehicles, entities must report crashes if the system was active within 30 seconds of the event and resulted in property damage or injury. L2 reporting is triggered by more severe outcomes, such as a fatality, an airbag deployment, or a collision involving a vulnerable road user. This allows NHTSA to identify safety defects and emerging trends in new technologies, ensuring that manufacturers meet their statutory obligations to keep public roads safe.

This is an important piece of the database as it directly instructs the reporting process for crashes involving L2 vehicles. This ensures that the company remains compliant with the specific 2026 data-submission windows (required to report any incident involving serious injury or property damage within 1 to 10 days). By using this data source, the LLM can accurately advise "Incident Report" triggers that may differentiate between a standard fender-bender crash and a federally reportable ADAS failure.

3. Data Processing Workflow

Given that the main goal of the LLM is to ingest laws with minimal cleaning and support from humans, the data processing was extremely simple. Legislation and other regulatory documents were transformed into easily readable text files from reliable sources like the National Highway Traffic Safety Administration and the Pennsylvania General Assembly. The txt files were then added to a database of all other sources that the model can easily pull from.

4. Data Limitations

Due to the nature of text files, these data files can not self-update. This provides major limitations in terms of how long the data is relevant for. As previously mentioned in section 1.3, with the laws and regulations of the automotive vehicle industry constantly evolving, this can make old laws completely obsolete within a matter of days. However, for a Proof-of-Concept, the main focus is on how the model outputs the information in the data and less about the relevance of the data itself. In a full-scale model, a webcrawler would be utilized to ensure the most up-to-date information is always used by the LLM.

Additionally, there is significantly more data on L2 autonomous driving nationwide. Since the data used by this initial LLM had to be discovered and processed by hand, there were severe limitations due to the time constraint of this project, as well as computer storage capacity. Ultimately, these barriers forced the decision to focus on PA and only the most important laws and regulations surrounding L2 autonomous vehicles. A full-scale version of this would utilize a webcrawler to find all laws and regulations in every state, not just the critically important PA laws.

2.1.3 Demo

We implemented a Gradio-based interactive demo that allows our users to ask either legal or academic questions related to ADAS. The system will automatically classify the query, retrieve the most relevant documents from the vector database, and generate a grounded response with source attribution.

1. Figure 1: Demo Landing Page

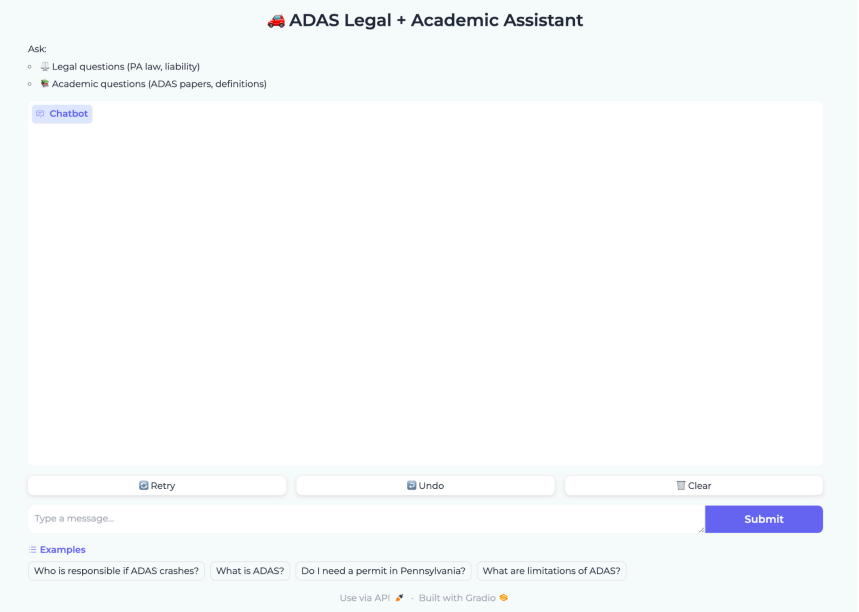
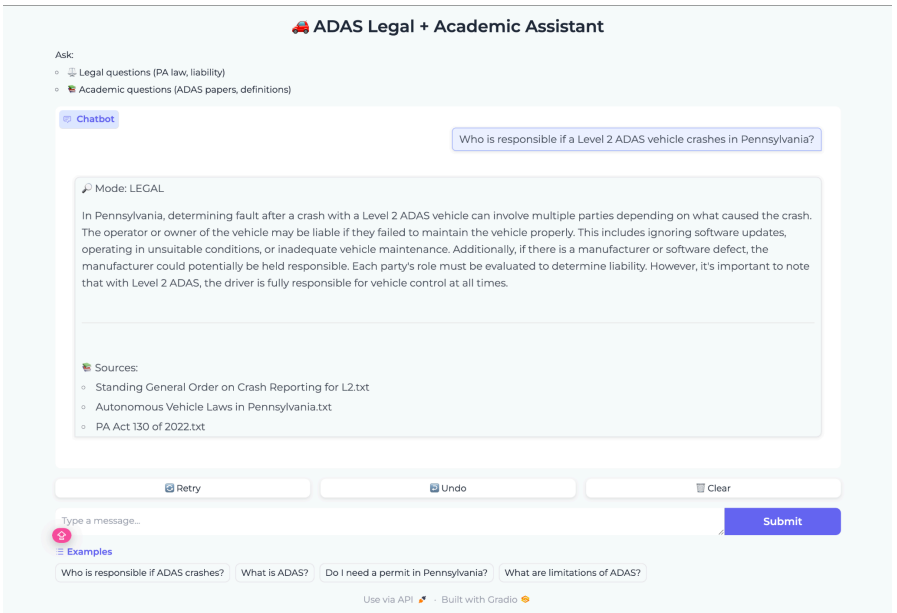


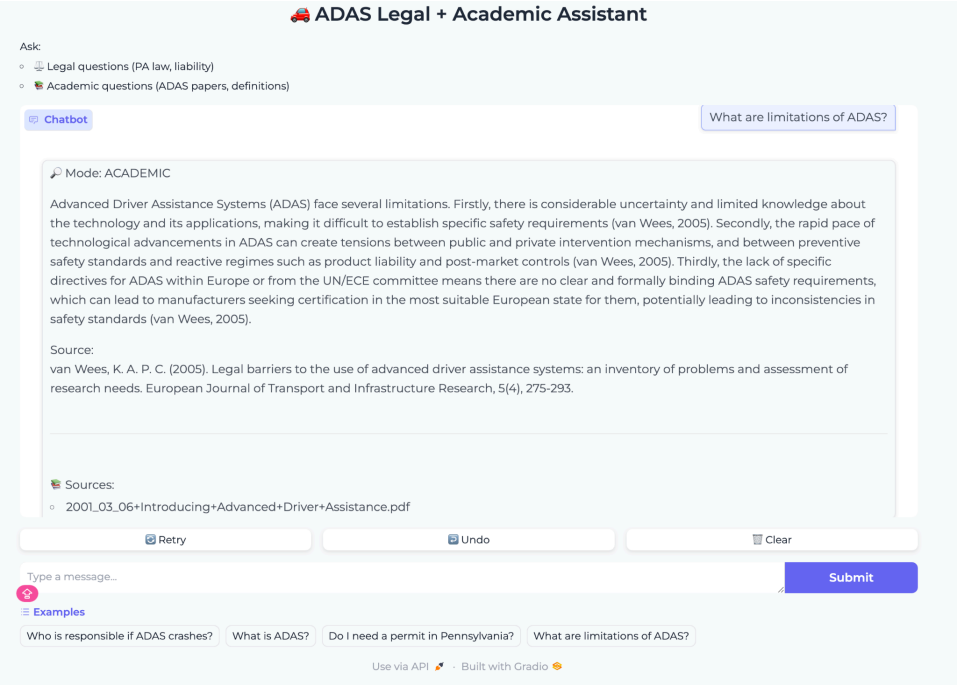
Figure 1 shows the user interface of the prototype. Users can enter any relevant questions and interact with the system through this lightweight chat interface.

2. Figure 2: Legal Query Example



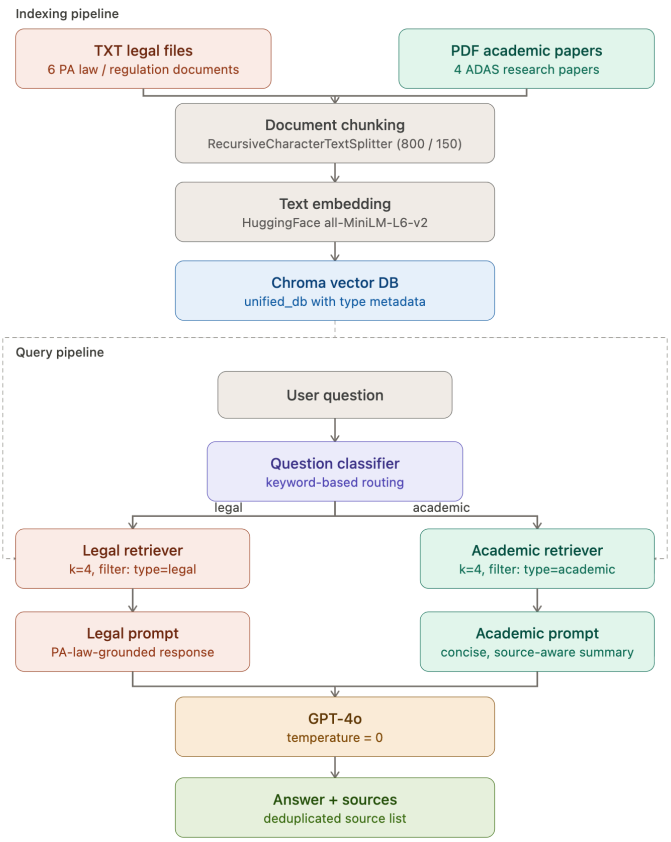
For legal queries, the system routes the question to the legal retrieval pipeline, which only searches in the Pennsylvania legal and regulatory database. The legal prompt constrains the model to answer only from the retrieved legal context and to abstain when evidence is insufficient. The output also includes the source documents used in retrieval, improving traceability and reducing hallucination risk.

3. Academic Query Example



For academic queries, the system retrieves from the academic paper database instead of the legal database. This enables the assistant to provide research-based definitions, background explanations, and discussion of broader ADAS limitations while keeping legal and academic evidence separated.

4. Workflow Diagram



The ADAS Legal-Academic Assistant is built on a dual-pipeline RAG architecture designed to separate legal reasoning from academic inquiry. During indexing, Pennsylvania regulatory documents and ADAS academic papers are chunked, embedded using HuggingFace's all-MiniLM-L6-v2, and stored in a Chroma vector database with type metadata tags that preserve the boundary between the two corpora. At query time, a keyword-based classifier routes each user question into either a legal or academic retrieval path, fetching the four most relevant chunks via metadata-filtered search. Each path uses a dedicated prompt template — the legal prompt grounds responses strictly in Pennsylvania law and encodes that the human driver remains legally responsible under Level 2 ADAS, while the academic prompt prioritizes concise, source-aware summaries. Both paths feed into GPT-4o at temperature 0, and the final output includes a list of retrieved source files for full traceability.

2.2 Product Advantages

2.2.1 Technical Advantages

The proposed system achieves strong technical robustness through a tightly controlled Retrieval-Augmented Generation (RAG) pipeline that minimizes hallucination and enforces domain-specific reasoning. Specifically, user queries are first classified into predefined categories (legal vs. academic) using a lightweight keyword-based classifier. This classification step enables routing queries to separate retrieval pipelines, each backed by a vector database constructed using sentence-transformer embeddings (all-MiniLM-L6-v2) and stored in a persistent Chroma index. By applying metadata-based filtering during retrieval, the system ensures that only contextually relevant documents are retrieved, effectively reducing noise and preventing cross-domain contamination.

To further enhance factual reliability, the system adopts a strict grounding mechanism at the prompt level. The legal assistant is explicitly instructed to generate responses solely based on retrieved statutory content and to abstain from answering when sufficient evidence is not available. This design directly mitigates the hallucination behavior commonly observed in general-purpose LLMs. Additionally, the retrieval process employs chunking strategies (chunk size = 800, overlap = 150) to balance semantic completeness and retrieval granularity, improving both recall and contextual coherence in downstream generation.

Another critical advantage lies in the use of a restricted and curated dataset. Unlike open-domain LLMs trained on heterogeneous internet data, this system operates on a controlled corpus consisting of Pennsylvania-specific traffic laws, HAV regulations, and selected academic literature on ADAS and liability. This restriction not only enhances interpretability and controllability but also enables the enforcement of domain constraints, such as consistently reflecting Level 2 automation principles where the human driver retains full responsibility. The separation of legal and academic sources also allows the system to adapt its response style and reasoning depth depending on the query type, improving both precision and usability.

Finally, the architecture is inherently extensible and production-oriented. The modular design—comprising document loaders (TXT/PDF), embedding generation, vector storage, retrieval filtering, and prompt templating—allows seamless integration of additional jurisdictions, regulatory updates, or technical literature without retraining the underlying language model. This makes the system particularly suitable for real-world deployment in ADAS engineering workflows, where regulatory compliance, traceability, and up-to-date

knowledge are critical.

2.2.2 Key Benefits

The proposed system delivers substantial operational benefits by significantly improving the speed, structure, and usability of regulatory knowledge access for ADAS engineering workflows. First, compared to traditional legal consultation processes, which often require manual interpretation of lengthy statutes or consultation with legal experts, the system provides near real-time responses through an automated RAG pipeline. By leveraging pre-indexed regulatory documents and efficient vector similarity search, relevant legal context can be retrieved and synthesized within seconds, dramatically reducing the latency between question formulation and actionable insight generation.

Second, the system offers a higher level of structural clarity compared to conventional engineering documentation. While engineering documents and regulatory texts are typically fragmented, verbose, and difficult to navigate, the model transforms unstructured source materials into concise, context-aware answers. This is achieved through prompt-controlled generation that enforces domain-specific response formats and prioritizes relevant evidence. Additionally, the separation of legal and academic knowledge sources allows the system to deliver responses that are both technically grounded and appropriately scoped, avoiding ambiguity that often arises in mixed or poorly structured documentation.

Finally, the system is designed to support real-time decision-making in safety-critical and compliance-sensitive scenarios. By integrating retrieval, reasoning, and response generation into a unified pipeline, engineers can query regulatory requirements, liability boundaries, and technical constraints during system design or testing phases without interrupting their workflow. The deterministic configuration of the language model further enhances consistency and reproducibility, which are essential for engineering validation and auditability. As a result, the system functions not only as a knowledge assistant but also as a decision support tool that bridges the gap between regulatory interpretation and engineering implementation.

2.3 Key Innovations

Our project introduces several key innovations that differentiate it from a general-purpose LLM application, with a total of three key points listed below:

- Scoped Legal Dataset Design: We adopt a scoped legal dataset design, where the model focuses on a well-defined subset of regulations—specifically, state laws related to SAE Level 2 driving—rather than attempting to cover all legal domains. By narrowing the scope, the model can achieve higher relevance and accuracy when interpreting regulatory content. This design reduces noise from unrelated legal domains and ensures that outputs remain focused on ADAS-specific requirements, making the system more practical and reliable for engineering use.

- Convert Laws to Behavior Mapping: Our system goes beyond traditional legal analysis by implementing a law-to-behavior mapping framework, which directly translates regulatory language into structured ADAS behavioral requirements. Instead of generating general summaries, the model identifies key elements such as conditions and required actions, and converts them into engineering-relevant

requirements. This enables developers to more easily integrate regulatory constraints into system design, effectively bridging the gap between legal interpretation and technical implementation.

- Human-in-the-loop Compliance Framework: Our approach incorporates a human-in-the-loop compliance framework. The model is designed to support engineers by providing structured and traceable outputs, while leaving final validation and decision-making to human experts. This ensures that outputs are reviewed for accuracy and compliance before being applied, reducing the risk of errors and improving system accountability. Rather than replacing human judgment, the system enhances efficiency while maintaining necessary oversight.

2.4 Core Competitive Advantage

The core competitive advantage of the proposed system lies in its unique combination of structured regulatory data, high model controllability, and deep industry specialization in the ADAS domain.

First, the system is built upon a curated and structured regulatory knowledge base, rather than relying on unfiltered, open-domain data. By organizing legal documents (e.g., Pennsylvania traffic laws and HAV regulations) and academic literature into a unified yet metadata-tagged corpus, the system enables precise and context-aware retrieval. This structured approach transforms inherently complex and fragmented regulatory texts into machine-interpretable knowledge, creating a data advantage that is both domain-specific and difficult to replicate without significant curation effort. Moreover, the architecture is inherently extensible: the system can be easily customized by incorporating additional regulatory datasets from different U.S. states or jurisdictions. As new legal documents are collected, processed, and embedded into the vector database, the model can seamlessly expand its coverage without requiring retraining. This enables scalable deployment across multiple regions while maintaining consistent performance and interpretability, effectively turning data accumulation into a long-term competitive moat.

Second, beyond leveraging existing academic literature for definitions and explanations, the system is capable of integrating proprietary, organization-specific knowledge through a structured knowledge injection process. In practice, domain experts, legal professionals, and industry executives can collaboratively define key concepts, responsibilities, and operational interpretations (e.g., ADAS liability boundaries or driver monitoring expectations). These internally developed definitions are then formalized into structured reports and incorporated into the system's knowledge base. As a result, the model is not limited to generic or externally sourced definitions, but instead reflects customized, high-confidence interpretations aligned with real-world industry practices. This creates a differentiated "definition layer" that enhances both consistency and strategic alignment across users.

Third, the system provides strong model control capabilities, addressing one of the key limitations of general-purpose LLMs—namely, their "black-box" nature. Through explicit architectural design choices, including query classification, metadata-filtered retrieval, prompt constraints, and deterministic generation settings (e.g., temperature = 0), the system ensures that outputs are traceable, auditable, and bounded within predefined knowledge scopes. This

level of control is particularly critical in safety- and compliance-sensitive applications, where explainability and reliability are more important than generative flexibility.

Finally, the system achieves a high degree of industry adaptation by focusing specifically on the ADAS vertical. Unlike horizontal AI assistants, this solution embeds domain-specific logic—such as Level 2 driver responsibility frameworks and regulatory interpretation rules—directly into its reasoning pipeline. This vertical integration allows the system to bridge the gap between legal requirements and engineering implementation, making it not only a knowledge retrieval tool but also a domain-aware decision support system. As a result, the solution offers a defensible competitive position by combining proprietary data structuring, controlled AI behavior, and deep alignment with real-world ADAS engineering needs.

2.5 Technical barriers

Our proposed system also faces several technical barriers that may impact scalability and adoption.

First, there are high costs associated with organizing and maintaining regulatory data. Legal documents are often unstructured, fragmented, and continuously evolving. Preparing these datasets for use in a retrieval-augmented system requires significant effort in data cleaning, structuring, labeling, and ongoing updates to ensure accuracy and compliance.

Second, the prompt-and-retrieval (RAG) design introduces implementation complexity. Building an effective system requires careful coordination between data preprocessing, embedding strategies, retrieval quality, and prompt engineering. Achieving reliable and consistent outputs is not straightforward, and the system design is difficult to replicate without iterative experimentation and domain-specific tuning.

Finally, there is a high barrier to entry due to industry-specific knowledge requirements. Developing a system that correctly interprets legal language and translates it into ADAS-relevant behaviors requires expertise in both regulatory frameworks and automotive engineering. This combination of interdisciplinary knowledge makes it challenging for newcomers to build contributing solutions without substantial domain experience.

3、Future System Vision

3.1 Capability Evolution

- Expansion from a single-state regulatory framework (Pennsylvania) to the integration of multi-state and federal regulatory systems:

The current proof-of-concept LLM only covers Pennsylvania laws and regulations governing autonomous vehicles on and off the highway. In a final version of the model, all 50 states would be covered to ensure compliance across the entire nation.

As previously mentioned, a current critical problem faced by the autonomous driving industry is that regulations vary state by state, causing compliance issues when driving over state lines. Including all 50 states completely solves this critical problem, as the LLM's outputs can be

adjusted to always follow the strictest state guidelines, ensuring that all state legislatures are followed at all times.

- Expansion from Level 2 ADAS to compliance support for Level 3–4 autonomous driving systems:

Currently, the LLM is focused on only L2 autonomous vehicles, with information about L3 and L4 only to ensure the LLM's output is specific to the L2 ADAS requirements. However, this LLM can be easily changed to focus on all levels of autonomous vehicles. By implementing documentation surrounding all levels of autonomous vehicles into the database as well as changing the prompt of the LLM from "L2 = driver fully responsible" to "L3 = driver must be present and available to take over within a few seconds when requested by the system" or "L4 = vehicle can drive itself entirely without human intervention", a final version of this LLM can be updated to satisfy any SAE driving level needed by the user.

- Evolution from static regulatory retrieval to dynamic regulatory updates and real-time synchronization mechanisms:

As previously mentioned, one of the most critical problems faced by the autonomous driving industry is that both state and federal laws and regulations are constantly evolving with the advancement of technology, leaving a database filled with only text files and PDFs obsolete after only a few days. To solve this issue, a webcrawler can be implemented into the database, thus providing for continuous updates from reliable websites without a developer or legal person having to update a static file by hand. This allows any autonomous vehicle system to never worry about an old regulation being outdated, as it is constantly being updated within the database.

3.2 Functional Roadmap

Our system is designed to evolve through four phases, each expanding its capability from legal understanding to engineering-level decision support and validation.

Phase 1: Legal Understanding

In the first phase, the system focuses on accurate retrieval and interpretation of regulatory text within a defined scope, specifically SAE Level 2 driving laws. By leveraging a structured legal dataset and retrieval-augmented generation (RAG), the model can identify relevant legal clauses and provide correct interpretations. This phase establishes the foundation for ensuring that all outputs are traceable to authoritative legal sources.

Phase 2: Compliance Analysis and Risk Identification

Building on legal understanding, the second phase enables the system to perform compliance analysis for specific driving scenarios. The model evaluates whether a given ADAS behavior aligns with regulatory requirements and identifies potential compliance risks. For example, it can assess whether a driver monitoring strategy satisfies legal expectations or flag scenarios where system behavior may lead to regulatory violations.

Phase 3: Decision Support and Behavioral Recommendations

In the third phase, the system transitions from analysis to decision support, translating regulatory requirements into actionable ADAS behavioral recommendations. This includes generating structured outputs such as lane behavior rules, driver monitoring triggers, and warning conditions. The goal is to bridge the gap between legal language and engineering implementation, enabling developers to directly apply regulatory insights in system design.

Phase 4: Closed-Loop Validation System

The final phase introduces a closed-loop validation framework, where the system supports engineering validation and continuous improvement. This includes generating test scenarios, validating system behavior against regulatory requirements, and incorporating feedback into the model. By closing the loop between regulation, implementation, and validation, the system enhances reliability, traceability, and compliance over time.

4、Risk

4.1 Technical Risks

One important technical risk in this project is LLM hallucination. Even though the system is built on a controlled RAG pipeline with query classification, metadata filtered retrieval, prompt constraints, and deterministic generation, there is still a possibility that the model may produce statements that sound correct but are not fully supported by the retrieved legal text. This risk is especially important in the ADAS setting because the system is designed for compliance-sensitive use, where even a small unsupported claim could lead to incorrect engineering or legal assumptions. The challenge is heightened by the fact that ADAS technologies are complex, safety-critical, and often difficult to evaluate with simple rules alone.

A second technical risk is the misinterpretation of regulations. The project focuses on Pennsylvania law and Level 2 ADAS-related materials, but the legal meaning of compliance is still not always straightforward. The uploaded sources show that ADAS and higher automation systems create complicated questions about driver responsibility, manufacturer responsibility, warning design, and system limitations. In addition, the NHTSA Standing General Order itself requires careful interpretation, since Level 2 reporting depends on specific crash conditions such as fatalities, air bag deployment, vulnerable road users, or hospital transport. Because legal texts often depend on precise wording, context, and edge cases, the system may retrieve relevant material but still oversimplify how a rule applies to a real scenario. This is particularly important for Level 2 systems, where the human driver must remain engaged, but liability questions may still become more complex when system behavior, driver behavior, and product design interact.

4.2 Operation Risks

One major operational risk is the misuse of model results by users. The purpose of the system

is to support compliance understanding, not to replace legal review or act as a final legal authority. However, users may treat the output as a complete answer and rely on it without checking the sources or considering whether the scenario falls outside the system's scope. This risk is realistic in the ADAS context because prior literature repeatedly shows that users may misunderstand the limits of assistance systems, overtrust automated behavior, or ignore warnings about proper supervision. A similar issue can occur with the model itself: even if the output is intended as decision support, users may apply it too confidently in design or compliance decisions. For that reason, the project should be framed as a human-in-the-loop support tool rather than a fully autonomous compliance engine.

Another operational risk is delays in regulatory updates. The current proof of concept relies on manually collected and uploaded legal and regulatory files, which means the database does not update automatically. This creates a real limitation because ADAS-related laws, reporting requirements, and state-level rules can change over time. The project outline already notes that static text files can become outdated quickly and that a full-scale version would need a web crawler or another update mechanism to keep the legal database current. If new rules are issued but not added to the system in time, the model may return answers that are technically grounded in the existing database but no longer reflect the most current regulatory environment. In a compliance-focused application, outdated information can be almost as risky as incorrect information.

5. Appendix

5.1 References:

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